

IC Design of 2Ms/s 10-bit SAR ADC with Low Power

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Abstract

This paper presents the development of a 2Ms/s 10-bit very low-power CMOS SAR ADC which is realized in a 0.35 μm CMOS process. The design combines a capacitor array DAC, a dual-cross coupled pair comparator, and SAR digital logic to create 8 effective bits while consuming less than 3mW with a 3.3V power supply.

1. Introduction

The increasing sophistication of System-on-Chip (SOC) architectures comes an increasing need for low power analog to digital converter (ADC). The primary advantage of successive approximation register (SAR) ADCs is that they preferred to pipeline or sub-range ADCs due to their low power consumption, high resolution and accuracy, and smaller area. Because of these benefits and a good tradeoff between resolution and speed, SAR ADCs can often be integrated with other larger function blocks. [1,2]

2. Circuit Description

The Successive Approximation ADC usually consists of a sample-and-hold stage (S/H), a digital-to-analog converter (DAC), a comparator and a successive approximation register, and it basically implements a binary search algorithm. [1]

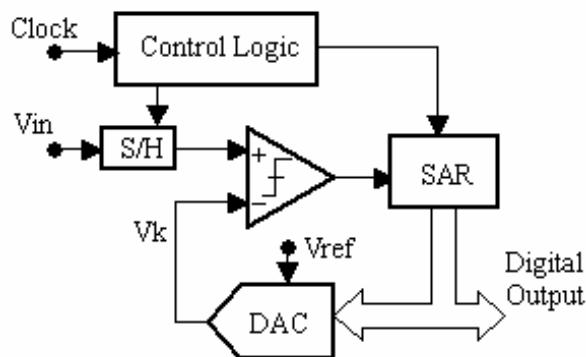


Fig.1. Block Diagram of basic SAR ADC.

The principle schematic of the proposed SAR-ADC is shown in Fig. 1. It depends on using the same hardware in N conversion cycles, where N is the resolution of the ADC. The analog input voltage (V_{in}) is held on a track/hold. V_{ref} is the reference voltage provided to the ADC. The successive approximation register is first set to mid-scale while the most significant bit (MSB) is set to '1'. This forces the DAC output to be $V_{ref}/2$, where a comparison is then performed to determine if V_{in} is less than or greater than V_k . If V_{in} is greater than V_k , the comparator outputs '1' and the MSB of

the N-bit register remains at logic '1'. Conversely, if V_{in} is less than V_k , the comparator outputs '1' and the MSB of the register is cleared to logic '0'. The SAR control logic then moves to the next bit, forces that bit high, then does another comparison. The sequence continues all the way down to the least significant bit (LSB) to complete the conversion, and the N-bit digital word is available in the register.

Generally speaking, an N-bit SAR ADC will require N comparison periods and will not be ready for the next conversion until the current one is completed. The entire conversion process is performed in three steps: the sample mode, the hold mode, and the redistribution mode.

2.1 DAC Design

The DAC is the most critical component of the SAR, and it is difficult to design. The SAR ADC's speed is mainly limited by the settling time of the DAC, which must settle to within the resolution of the overall converter, for example, $1/2\text{LSB}$, and the comparator, which must resolve small differences in input voltage within the specified time[3].

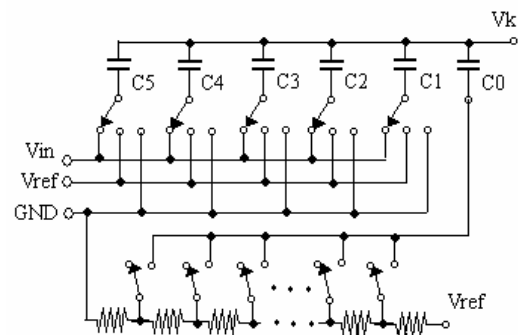


Fig.2. DAC Topology.

Table 1. Capacitor values

Capacitor	Value	Unit
C0	200	fF
C1	200	fF
C2	400	fF
C3	800	fF
C4	1.6	pF
C5	3.2	pF

The DAC utilizes a capacitor array and a resistive ladder to get a better performance shown in Fig.2. This architecture is special because it reduces the size of the switches by

The capacitor and resistor switches were controlled by outputs from the control logic. The switches and capacitors used in the capacitor array have to be optimized so that the charge applied to the capacitors settles in the required time; otherwise, it will cause errors in the conversion period. During sampling, V_{in} is applied to the cap array. In the conversion phase, the voltage on the top plate of the capacitors also can be treated as a time constant. Assuming that the voltage source has negligible series resistance these two time constants are roughly equal and can be given as

$$\tau_P = \text{RONCT} \quad (1)$$

Where R_{ON} is the resistance of the switch and C_T is the total capacitance of the array.

The MSB capacitor has the most deleterious effect on the signal-to-noise and distortion ratio (SNDR). If the mismatch affects only the MSB capacitor, the deviation has to be much lower than 1% to limit the loss in SNDR to 3dB.

2.2. Comparator Design

The requirements of the comparator are speed and accuracy. The requirements for the comparator are greatly affected by parasitic capacitances. Comparator offset affects the accuracy of the conversion, but it does not affect overall linearity, as it appears as an offset in the overall transfer characteristic. The offset will be stored on the capacitor array and is subsequently cancelled. Additionally, the comparator needs to resolve voltages within the accuracy of the overall system, and needs to be as accurate as the overall system.

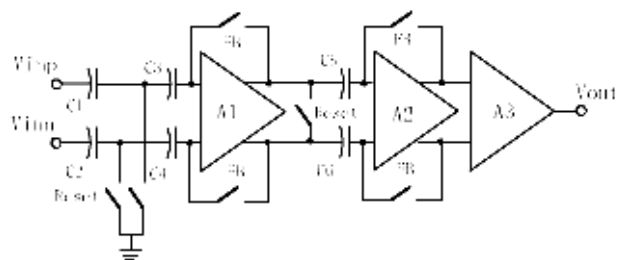
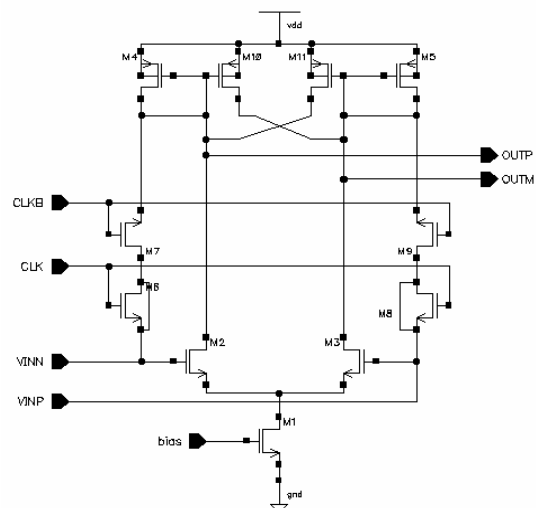


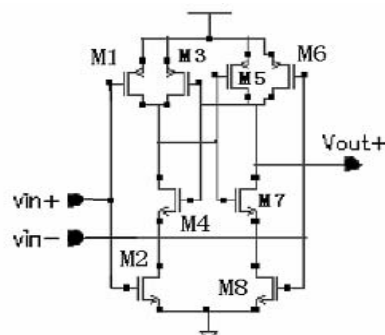
Fig.3. The comparator architecture.

The paper presents a designed comparator that consists of three stages to complete the operation, and each stage of the comparator is controlled by the clock. Switching the clock produces distortion on the input signal. Clock feed through effect is arising with the parasitic capacitance of the

transistors. To solve this problem, coupling capacitor is set between two stages. When the switch FB and Reset are turned on, each stage is reset to cancel offset error. Thus, the input offset as well as the output offset is decreased and have much less influence on the accuracy. The topology of the first and second stage is shown in Fig. 4, with the output stage of the comparator.



(a) The first stage of comparator



(b) The output stage of comparator

Fig.4. Comparator Topology

The topology of the comparator is shown in Fig.4. In general, this architecture consumes less power than other comparator topologies and is a good trade-off between speed and power for the design. This comparator only consumes a maximum of 2mW of power in the worst case with approximately 12-bit conversion accuracy.

2.3 Control Logic

In the sampling phase the comparator is short-circuited and, therefore, the reference voltage V_m , which is used for simplifying the mid-voltage between the power supply voltage and ground, rises at the comparator input. Thus, the capacitor array is charged to the difference of the reference voltage V_m and the input voltage V_{in} . During the converting phase, the binary weighted capacitors are either connected to a positive or a negative reference voltage according to the bits of the control register [3,4].

A non-overlapping two-phase clock is generated from the overall sampling clock provided by the SAR to control the sequence of the sample and hold. In the comparator, both signals are provided in complementary form to control the n-MOS switches and the related n-MOS dummy switch devices to reduce the influence of the charge injection. The operating point of the switch transistors determines the minimum settling time.

The control logic was implemented using a double shift-register based architecture that has the advantage of minimizing active power at the expense of increased area and standby current. So the digital logic and DAC consume very minimal power.

2.4 Simulation and Experimental Results

With the tool of Spectre, simulation is performed to evaluate the characters of the ADC.

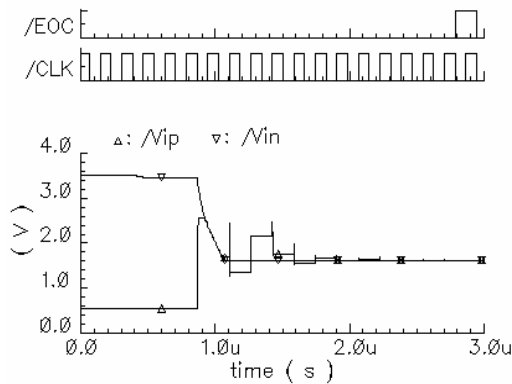


Fig.5. The transient simulation result.

Fig.5 show the transient simulation result of the SAR ADC. The signal end-of-convert (EOC) declares the end of the conversion. CLK is the operation clock. The two inputs of the comparator are Vip and Vin. It is clear enough that Vip successive approximates Vin. The conversion result is ready at the positive edge of EOC.

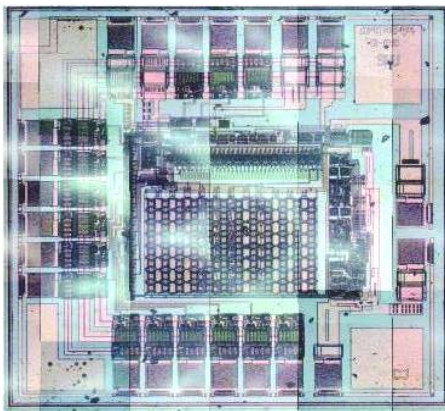


Fig.6. The photograph of ADC

A prototype of this ADC was designed and fabricated in a standard $0.35\mu\text{m}$ n-well CMOS process with single poly, four metal layers and a Metal-Insulator-Metal (MIM) capacitor option. Special attention was paid to the capacitive DAC and careful arrangement of the wiring makes the parasitic

capacitances also binary weighted. A chip photograph is shown in Fig. 6. Chip area is $0.9\text{ mm} \times 2$.

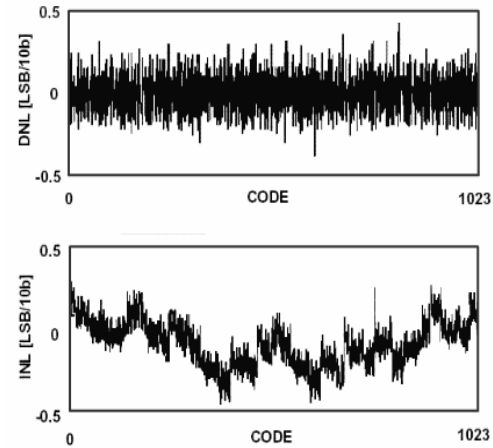


Fig.7. Measured INL and DNL at VDD = 3.3V

Measurements have been taken to get ADC's static characters. Exemplary plots of the differential and integral non-linearity (DNL and INL, respectively) are shown in Fig.7. It shows that the INL and DNL are less than 0.5LSB .

Conclusions

A low power single ended 2Ms/s 10-bit SAR A/D converter was designed in a $0.35\mu\text{m}$ CMOS technology using standard threshold devices. The overall area of the circuit is 0.4mm^2 . The resistor used in the DAC is implemented using a poly resistor. With a power dissipation of 3mW from a 3.3V supply, this ADC is suited for low power operation. In the future, there are still many ways to improve the available sample rates and reduce the chip size.

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